Constrained sphere packing and the interpolation problem for positive definite functions

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Joint work with Sujit Sakharam Damase, IISc, Bangalore

The team

Sujit (rising 4th year graduate student at IISc) will be visiting at MIT next year starting in the Fall. (Thanks to Jesse's notes for teaching him basic operator algebras.)

Acknowledgments

We would like to thank:

- The Indian Institute of Science, Bangalore
- The Indian Statistical Institute, Bangalore
- J C Bose Fellowship JCB/2021/000041 of SERB
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Thanks to Jeff Remmel (1948-2017)
for teaching me representation theory.

Permanently building the network in all areas with an understanding of the transient nature of political entities is key to a continuing human role in mathematical progress.

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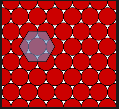
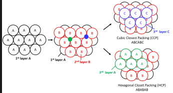
Lagrange



$n = 3$

Hales (90s)



$n = 8, 24$

Viazovska et al. (2010s)

E_8,

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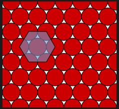

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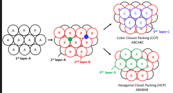

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For no other cases is the optimal density known exactly, although the optimal packings do not come from lattices.

Spherical codes

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A **spherical code** is a set of points in the the sphere in n real dimensional space. We say it has **minimum angle** to be the smallest θ such that every pair of distinct points in the code makes an angle of at least θ with respect to the origin.

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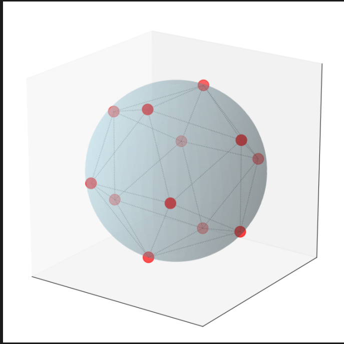

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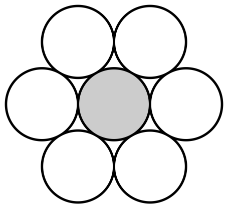

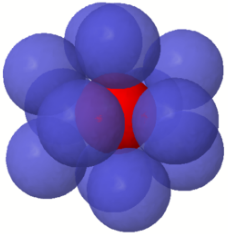

The kissing number is equal to the maximal size of a spherical code with minimum angle $\pi/3$.

Delsarte's method

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Observation: Let $\pi/2 < \theta < \pi$. Any spherical code with minimum angle θ has cardinality less than

$$\frac{\cos\theta - 1}{\cos\theta}.$$

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Strategy: Given $n \in \mathbb{N}$ and $0 < \theta < \pi/2$ find the maximum $\tilde{\theta} > \pi/2$ such that there is a map from the sphere in n-dimensional space to the sphere in a larger dimensional space such that every spherical code with minimum angle θ is mapped to a code with minimum angle at least $\tilde{\theta}$.

Positive definite functions

Positive definite functions

Let G be group, let H be subgroup of G. A (completely) positive definite function

$$f : G/H \to \mathbb{R} \text{ (resp. self-adjoint operators)}$$

satisfies: $[f(g_i^* g_j)]_{i,j}$ is positive semidefinite for all $g_1, \ldots, g_d \in G$.

Stinespring factorization

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Given a positive definite function $f : G/H \to \mathbb{R}$ it has a **Stinespring factorization**

$$f(g) = v^* \pi(g) v$$

where π is a representation of G and v is a vector which is invariant for $\pi|_H$. If f is continuous and G is a compact group, the Peter-Weyl decomposition gives this is a direct sum of irreducibles. (convex combination)

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The homomorphic analogue of **Tomiyama's theorem** gives that f is in fact bi-invariant,

$$f(h_1 g h_2) = f(g), \forall h_1, h_2 \in H, g \in G.$$

Schoenberg's theorem on positive definite functions

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If f is positive definite and continuous, $f(g) = \sum_{i=1}^{\infty} a_i C_i^{(n)}(\gamma_{11})$ where $a_i \geq 0$ and the $C_i^{(n)}$ are Gegenbauer polynomials, a natural basis for bi-invariant functions. (We will normalize to have $C_i^{(n)}(1) = 1$)
(Schoenberg '42, Bochner '41)

Operations on positive definite functions

Operations on positive definite functions

Given $C_i^{(n)}(\gamma_{11}) = v_i^* \pi_i(\gamma) v_i$, we see that

$$C_i^{(n)} C_j^{(n)} = \sum_{k=0}^{i+j} a_k C_k^{(n)}$$

where $\sum a_k = 1, a_k \geq 0$ as

$$C_i^{(n)}(\gamma_{11}) C_j^{(n)}(\gamma_{11}) = (v_i \otimes v_j)^* (\pi_i \otimes \pi_j)(\gamma)(v_i \otimes v_j)^*$$

and $\pi_i \otimes \pi_j$ may again be broken into irreducibles.

Kernel embedding

Kernel embedding

Theorem (Aronszajn's theorem)

Let X be a set. Let $A : X \times X \to \mathbb{C}$ such that $[A(x_i, x_j)]_{i,j}$ is positive semidefinite for arbitrary finite collections $x_1, \ldots, x_d$.
There exists a Hilbert space $\mathcal{H}$ and $\iota_x : X \to \mathcal{H}$ such that $\langle \iota_x, \iota_y \rangle = A(x, y)$.

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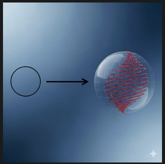
Thus a positive definite function $f : [-1, 1] \to \mathbb{R}$ such that $f(1) = 1$ induces an embedding via the kernel $f(\langle x, y \rangle)$ which lands in the sphere of the Hilbert space as $f(\langle x, x \rangle) = 1$.



Delsarte's method

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Theorem (Delsarte '72)

Let $0 < \theta < \pi$. Let $g : [-1, 1] \to \mathbb{R}$ be a positive definite function on the sphere in n dimensions such that $g(1) = 1$ and $g(x) \leq 0$ if $x \leq \cos\theta$.
Let $\bar{g}$ the constant coefficient of g in its Gegenbauer expansion. (That is the average value of $g(\langle x, y\rangle)$ over x, y in the unit sphere. Let $\mathcal{C}$ be spherical code with minimal angle θ. Then,

$$|\mathcal{C}| \leq \frac{1}{\bar{g}}.$$

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Linear programming methods: Delsarte-Goethals-Seidel, '77 Kabitiansky-Levenshtein '78.

Kissing number in $\mathbb{R}^2$

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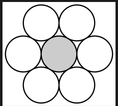

The polynomial

$$p(t) = \frac{4}{9}(t - 1/2)(t + 1/2)^2(1 + t)$$

satisfies

$$p(x) = \frac{1}{6}C_0^{(2)}(x) + \frac{1}{3}C_1^{(2)}(x) + \frac{5}{18}C_2^{(2)}(x) + \frac{1}{6}C_3^{(2)}(x) + \frac{1}{18}C_4^{(2)}(x)$$

$$\text{i.e. } p(\cos\theta) = \frac{1}{6} + \frac{1}{3}\cos\theta + \frac{5}{18}\cos 2\theta + \frac{1}{6}\cos 3\theta + \frac{1}{18}\cos 4\theta$$

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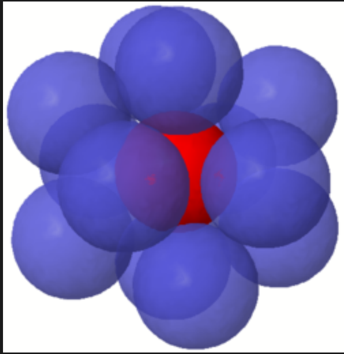

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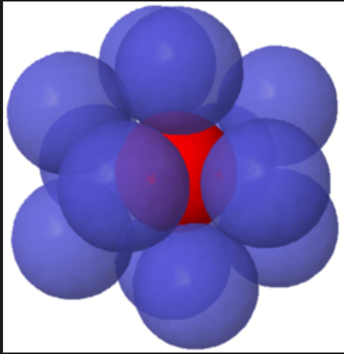

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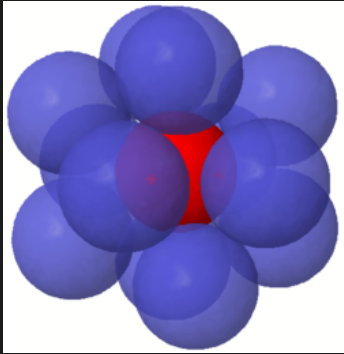

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Delsarte sharpness

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where $\bar{g}$ is the spherical average of g.

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Sharp	Not Sharp
$n = 2$ kissing number	$n = 3, 4$ kissing number
	Musin
$n = 8, 24$ kissing number	$X = \{-1, 1/2, -1/2, 1\}, n = 10$
Sloane, Odlyzko, 79	Damase, P. ($\frac{1}{\bar{g}} = 46.$)

Cohn-Elkies method

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Theorem (Cohn-Elkies, '03)

Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is an admissible function, is not identically zero, and satisfies the following two conditions:

1. $f(x) \leq 0$ *for* $|x| \geq 1$, *and*
2. $\hat{f}(t) \geq 0$ *for all* t.

Then the center density of n-dimensional sphere packings is bounded above by

$$\frac{f(0)}{2^n \hat{f}(0)}.$$

Constrained angle codes

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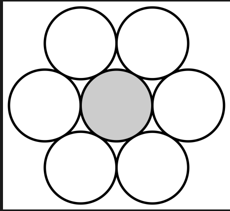

For example, the spherical code given by six equispaced points on the circle has cosines in $\{0, -1, 1/2, -1/2\}$.

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c.f. Krein, Jorgensen, etc. on similar but different interpolation problems. ($\Gamma - \Gamma$, not necessarily symmetric, only ask for f to induce a kernel on Γ.)

Arveson extension and pathological Class I representations

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Theorem (Damase-P.)

$SO(\infty)/1 \oplus SO(\infty)$ and $SO(2)/\{1\}$ are not pathological.

Continuous extension problem

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Let X be a finite subset of $[-1, 1]$. Let $f : X \to \mathbb{R}$. When is there a continuous positive definite function on the sphere in n-dimensions $\tilde{f} : [-1, 1] \to \mathbb{R}$ such that $\tilde{f}|_X = f$?

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Theorem (Damase, P.)

Let f correspond to a solvable continuous extension problem. There is a polynomial solution $\tilde{f}$.

Polynomial Delsarte sharpness

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Theorem (Nonantipodal sharpness)

Let $-1 \notin X$. *If* X *is Delsarte sharp, then there exists a positive definite polynomial* g *with* $\frac{1}{\bar{g}} = |C|$.

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Viazovska et al. functions for Cohn-Elkies are modular functions. (Low algebraic complexity.)

Delsarte hallucinations

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- Consider $X = \{-1, 1/2, -1/2, 1\}, n = 10$. There is no corresponding constrained angle code, but we have $\frac{1}{g} = 46$ is an integer. Many such examples exist of **Delsarte hallucinations.**

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- On the other hand, the example $X = \{1/3, -1/3, 1\}$ has a spherical code with 10 points in the 5 dimensional sphere, but the sign pattern comes from a Petersen graph.

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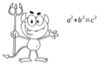
- On the other hand, the example $X = \{1/3, -1/3, 1\}$ has a spherical code with 10 points in the 5 dimensional sphere, but the sign pattern comes from a Petersen graph.
- Given an X, determining if there exists a subset of the n-sphere with cosines in X of some size k is a difficult combinatorial matrix completion problem.

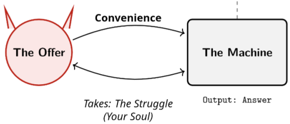
The future is now

Algebra is the offer made by the devil to the mathematician. The devil says: I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul: give up geometry and you will have this marvelous machine.



—*Sir Michael Atiyah*, 2002

$P(w_t|w_{<t}) = \text{softmax}(h_t)$



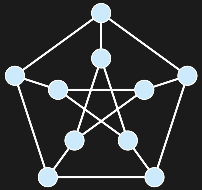
AI is the offer made by the devil to the mind.

The devil says: I will give you this powerful machine,
it will complete any thought you like.

All you need to do is give me your soul:
give up ***the struggle*** and you
will have this marvelous machine.

— Gemini, 2025

Delsarte Vision



The Petersen Graph

Seidel Matrix S

$$\begin{pmatrix} 0 & - & - & 1 & 1 & - & 1 & 1 & 1 & 1 \\ - & 0 & - & - & 1 & 1 & - & 1 & 1 & 1 \\ - & - & 0 & - & - & 1 & 1 & - & 1 & 1 \\ 1 & - & - & 0 & - & 1 & 1 & 1 & - & 1 \\ 1 & 1 & - & - & 0 & 1 & 1 & 1 & 1 & - \\ - & 1 & 1 & 1 & 1 & 0 & 1 & - & 1 & - \\ 1 & - & 1 & 1 & 1 & 1 & 0 & 1 & - & - \\ 1 & 1 & - & 1 & 1 & - & 1 & 0 & - & - \\ 1 & 1 & 1 & - & 1 & 1 & - & - & 0 & - \\ 1 & 1 & 1 & 1 & - & - & - & - & - & 0 \end{pmatrix}$$

Spectrum: $3^{(5)}, -3^{(5)}$

Gram Matrix G

Construct $X = \{1, \pm 1/3\}$:

$$G = I + \frac{1}{3} S \succeq 0$$

Discovery: AI = 爱

- Example suggested by **Gemini**.
- Verified & Formalized by **Damase & P.**

Soft Thresholding

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Theorem (Damase, P.)

The faithfulness constant of $[-\varepsilon, \varepsilon]$ *does not go to* 1 *if* $N \geq 4$.

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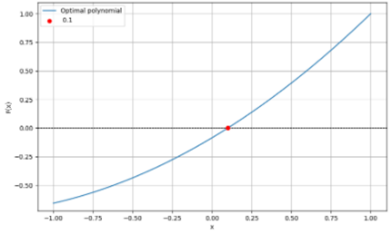

FIGURE 1. The thresholding function for $K = \{0.1\}, n = 3$